NYJC 2016 J2 H3 Math Preliminary Examination Mark Scheme

1(i)

$f'(x)=1-\frac{1}{1+x}=\frac{x}{1+x}>0$ for $0<x<1$ $\Rightarrow f\uparrow$ on $(0,1)$.

Moreover, $f(0)=0$. So for all $0<x<1$,

$$f(x)>0 \Rightarrow x-\ln(1+x)>0 \Rightarrow x>\ln(1+x).$$

(ii)

Reflecting $y=\ln(x+1)$ in the y-axis followed by the x-axis, we obtain $y=-\ln(1-x)$ and therefore we deduce that $-\ln(1-x)>x$.

Alternatively, define $g(x)=x+\ln(1-x)$ for $0<x<1$.

Then $g'(x)=1-\frac{1}{1-x}=-\frac{x}{1-x}<0$ since $0<x<1$. So $g\downarrow$ on $(0,1)$.

Since $g(0)=0$, we conclude that for all $0<x<1$,

$g(x)<0 \Rightarrow x+\ln(1-x)<0 \Rightarrow -\ln(1-x)>x$.

(iii)

Combining the results in **(i)** and **(ii)**, we have for $0<x<1$,

$\ln(1+x)<x<-\ln(1-x)$.

Put $x=\frac{1}{n}$, where $n\in\mathbb{Z}^+, n>1$ (so that $0<x<1$ $\because$ $0<\frac{1}{n}<1$), into above inequality gives

$$\ln\left(1+\frac{1}{n}\right)<\frac{1}{n}<-\ln\left(1-\frac{1}{n}\right)\Rightarrow \ln\left(\frac{n+1}{n}\right)<\frac{1}{n}<-\ln\left(1-\frac{1}{n}\right)$$

$$\Rightarrow \ln(n+1)-\ln n<\frac{1}{n}<\ln n-\ln(n-1)$$

for each $n\in\mathbb{Z}^+, n>1$.

Summing from $n = N$ to $3N$ gives

$$\sum_{n=N}^{3N}\left[\ln(n+1)-\ln n\right]<\sum_{n=N}^{3N}\frac{1}{n}<\sum_{n=N}^{3N}\left[\ln n-\ln(n-1)\right]$$

$$\begin{bmatrix} \ln(N+1)-\ln N \\ +\ln(N+2)-\ln(N+1) \\ \vdots \\ +\ln(3N+1)-\ln 3N \end{bmatrix}<\sum_{n=N}^{3N}\frac{1}{n}<\begin{bmatrix} \ln N-\ln(N-1) \\ +\ln(N+1)-\ln N \\ \vdots \\ +\ln 3N-\ln(3N-1) \end{bmatrix}$$

$$\Rightarrow \quad \ln(3N+1)-\ln N<\sum_{n=N}^{3N}\frac{1}{n}<\ln 3N-\ln(N-1)$$

$$\Rightarrow \quad \ln\left(\frac{3N+1}{N}\right)<\sum_{n=N}^{3N}\frac{1}{n}<\ln\left(\frac{3N}{N-1}\right)$$

(iv)

As $N\to\infty$, $\ln\left(\frac{3N+1}{N}\right)=\ln\left(3+\frac{1}{N}\right)\to\ln 3$ and

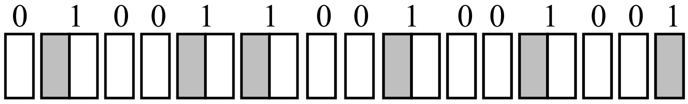
$\ln\left(\frac{3N}{N-1}\right)=\ln\left(3+\frac{3}{N-1}\right)\to\ln 3$

By squeeze theorem, $\sum_{n=N}^{3N}\frac{1}{n}\to\ln 3$.

2(a)	Let $I=\int_0^{\infty}\frac{\ln x}{x^2+a^2}\,\mathrm{d}x$. $x=\frac{a}{t}\Rightarrow \mathrm{d}x=-\frac{a}{t^2}\mathrm{d}t$ $I=\frac{1}{a}\int_{\infty}^{0}\frac{\ln\left(\frac{a}{t}\right)}{\left(\frac{a}{t}\right)^2+a^2}\left(-\frac{a}{t^2}\right)\mathrm{d}t=\frac{1}{a}\int_0^{\infty}\frac{\ln a-\ln t}{1+t^2}\,\mathrm{d}t$ $=\frac{1}{a}\left[\int_0^{\infty}\frac{\ln a}{1+t^2}\,\mathrm{d}t-\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t\right]$ $=\frac{\ln a}{a}\left[\tan^{-1}t\right]_0^{\infty}-\frac{1}{a}\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t$ $=\frac{\pi\ln a}{2a}-\frac{1}{a}\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t$ Thus we have proved the following: (*) $I=\int_0^{\infty}\frac{\ln t}{a^2+t^2}\,\mathrm{d}t=\frac{\pi\ln a}{2a}-\frac{1}{a}\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t$. Setting $a=1$ in (*) gives $\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t=\frac{\pi\ln 1}{2(1)}-\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t$ $=-\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t$ $\Rightarrow\ 2\int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t=0\ \Rightarrow\ \int_0^{\infty}\frac{\ln t}{1+t^2}\,\mathrm{d}t=0$ Hence $I=\frac{\pi\ln a}{2a}$
(b)	$\int_0^{\frac{1}{4}\pi}\tan^{n+2}x\,\mathrm{d}x+\int_0^{\frac{1}{4}\pi}\tan^{n}x\,\mathrm{d}x=\int_0^{\frac{1}{4}\pi}\tan^{n}x\left(1+\tan^2x\right)\,\mathrm{d}x$ $=\int_0^{\frac{1}{4}\pi}\tan^{n}x\sec^2x\,\mathrm{d}x$ $=\frac{1}{n+1}\left[\tan^{n+1}x\right]_0^{\frac{1}{4}\pi}$ $=\frac{1}{n+1}$
(b)(i)	Volume of solid $=\pi\int_0^{\frac{1}{4}\pi}\tan^4x\,\mathrm{d}x$ $=\pi\left(\frac{1}{3}-\int_0^{\frac{1}{4}\pi}\tan^2x\,\mathrm{d}x\right)$ $=\pi\left[\frac{1}{3}-\left(1-\int_0^{\frac{1}{4}\pi}1\,\mathrm{d}x\right)\right]$ $=\pi\left(\frac{1}{4}\pi-\frac{2}{3}\right)$

(b)(ii)	We have $I_{n+2}+I_n=\dfrac{1}{n+1}$ for $n=0,1,2,\ldots.$ (*) For $0<x<\dfrac{\pi}{4}$, $0<\tan x<1 \Rightarrow \tan^n x>\tan^{n+2}x \Rightarrow I_n>I_{n+2}$. Thus $\dfrac{1}{n+1}-I_n<I_n \Rightarrow I_n>\dfrac{1}{2(n+1)}$. Similarly, replacing n by $n-2$ in (*), $I_n+I_{n-2}=\dfrac{1}{n-1}$ for $n=2,3,4,\ldots.$ $I_n<I_{n-2}=\dfrac{1}{n-1}-I_n \Rightarrow I_n<\dfrac{1}{2(n-1)}$. Hence we have $\dfrac{1}{2(n+1)}<I_n<\dfrac{1}{2(n-1)}$ for $n>1$. Putting $n=200$, we have $\dfrac{1}{2(200+1)}<I_{200}<\dfrac{1}{2(200-1)} \Rightarrow 0.00249<I_{200}<0.00251$ Hence $I_{200}=0.0025$ to 4 decimal places.
3(i)	Consider the square $(x-y)^2$. $(x-y)^2\ge 0 \Rightarrow x^2+y^2-2xy\ge 0$ $\Rightarrow \dfrac{1}{2}(x^2+y^2)\ge xy$
(ii)	By **(i)**, $\sum_{i=1}^{n}A_iB_i\le\dfrac{1}{2}\sum_{i=1}^{n}\left(A_i^2+B_i^2\right)$ $=\dfrac{1}{2}\sum_{i=1}^{n}\left[\dfrac{a_i^2}{\sum_{i=1}^{n}a_i^2}+\dfrac{b_i^2}{\sum_{i=1}^{n}b_i^2}\right]$ $=\dfrac{1}{2}\left[\sum_{i=1}^{n}\left(\dfrac{a_i^2}{\sum_{i=1}^{n}a_i^2}\right)+\sum_{i=1}^{n}\left(\dfrac{b_i^2}{\sum_{i=1}^{n}b_i^2}\right)\right]$ $=\dfrac{1}{2}\left[\dfrac{\sum_{i=1}^{n}a_i^2}{\sum_{i=1}^{n}a_i^2}+\dfrac{\sum_{i=1}^{n}b_i^2}{\sum_{i=1}^{n}b_i^2}\right]$ $=\dfrac{1}{2}(1+1)=1$ Define $S_a=\left(\sum_{i=1}^{n}a_i^2\right)^{\frac{1}{2}}, S_b=\left(\sum_{i=1}^{n}b_i^2\right)^{\frac{1}{2}}$. Thus $A_i=\dfrac{a_i}{S_a}, B_i=\dfrac{b_i}{S_b}$.

	From the above result, we have $1 \geq \sum_{i=1}^{n} A_iB_i = \sum_{i=1}^{n} \left\{\left(\frac{a_i}{S_a}\right)\left(\frac{b_i}{S_b}\right)\right\} = \frac{1}{S_aS_b}\sum_{i=1}^{n} a_ib_i$ $\Rightarrow S_aS_b \geq \sum_{i=1}^{n} a_ib_i$ $\Rightarrow \left(\sum_{i=1}^{n} a_ib_i\right)^2 \leq S_a^{\,2}S_b^{\,2}$ on squaring $\Rightarrow \left(\sum_{i=1}^{n} a_ib_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^{\,2}\right)\left(\sum_{i=1}^{n} b_i^{\,2}\right)$ Equality is achieved if, for all $i = 1, 2, \ldots, n$, $A_i = B_i \Rightarrow \dfrac{a_i}{\left(\sum_{i=1}^{n} a_i^{\,2}\right)^{\frac{1}{2}}} = \dfrac{b_i}{\left(\sum_{i=1}^{n} b_i^{\,2}\right)^{\frac{1}{2}}}$ $\Rightarrow \dfrac{a_i}{b_i} = \dfrac{\left(\sum_{i=1}^{n} a_i^{\,2}\right)^{\frac{1}{2}}}{\left(\sum_{i=1}^{n} b_i^{\,2}\right)^{\frac{1}{2}}} = k =$ constant
(iii)	In any triangle ABC, $A + B + C = 180^\circ \Rightarrow \frac{A}{2} + \frac{B}{2} = 90^\circ - \frac{C}{2}$ $\tan\left(\frac{A}{2} + \frac{B}{2}\right) = \tan\left(90^\circ - \frac{C}{2}\right) = \cot\frac{C}{2} = \dfrac{1}{\tan\frac{C}{2}}$ $\Rightarrow \dfrac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \dfrac{1}{\tan\frac{C}{2}}$ $\Rightarrow \left(\tan\frac{A}{2} + \tan\frac{B}{2}\right)\tan\frac{C}{2} = 1 - \tan\frac{A}{2}\tan\frac{B}{2}$ $\Rightarrow \tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$ Define $a = \tan\frac{A}{2}, b = \tan\frac{B}{2}, c = \tan\frac{C}{2}$ and consider the two sequences of numbers a, b, c and b, c, a. Since $0 < \frac{A}{2}, \frac{B}{2}, \frac{C}{2} < 90^\circ$, we conclude that $a, b, c > 0$. By **(ii)**, $1 = ab + bc + ca \leq \left(a^2 + b^2 + c^2\right)\left(b^2 + c^2 + a^2\right) = \left(a^2 + b^2 + c^2\right)^2$ $\Rightarrow a^2 + b^2 + c^2 \geq 1$ $\Rightarrow \tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1.$
4(a)	Given a possible choice of 6 cards so that no two adjacent cards are selected,

	each of the first 5 chosen cards will be followed by at least one non-chosen card. We map each pair of chosen and following non-chosen card to a 1 and the 6^{th} chosen card itself to a 1 and the remaining 9 non-chosen cards to a 0. Thus such a choice of 6 cards corresponds to a binary sequence of length 15 and exactly 6 ones. Conversely, given a binary sequence consisting of 9 0's and 6 1's, map each 0 to a non-chosen card, each of the first 5 1's to a chosen card followed by a non-chosen card, and the last 1 to a chosen card. 0 1 0 0 1 1 0 0 1 0 0 1 0 0 1  By Bijection Principle, number of ways the 6 cards can be chosen so that no two adjacent cards are selected is equal to $\binom{15}{6} = 5005$.
(b)	T_n : number of ways cards can be chosen from *n* cards (including the event where no card is chosen) so that no two adjacent cards are selected. If the first card is chosen, then the second card cannot be chosen. Number of ways to choose from the remaining $n-2$ cards so that no two adjacent cards are selected is given by T_{n-2}. Otherwise, choose from the remaining $n-1$ cards so that no two adjacent cards are selected and the number of ways to do so is given by T_{n-1}. By the addition principle, $T_n = T_{n-1} + T_{n-2}$. $T_1 = 2$ and $T_2 = 3$ $T_3 = 2+3=5$, $T_4 = 8$, $T_5 = 13$, $T_6 = 21$
(c)(i)	Case 1: 1 friend gets 3 cards and the other 5 friends get 1 card each. Number of ways $= \binom{8}{3} 6! = 40320$ Case 2: 2 friends get 2 cards each and the other 4 friends get 1 card each. Number of ways $= \dfrac{\binom{8}{2}\binom{6}{2}}{2!} 6! = 151200$ By Addition Principle, total number of ways to distribute the 8 distinct cards among 6 friends $= 40320 + 151200 = 191520$. Alternative method: Total number of ways to distribute the 8 distinct cards among 6 friends $= S(8,6) \times 6!$

(c)(ii)

Number of ways the 8 distinct cards be distributed into 5 piles is given by $S(8,5)$.

<u>Case 1</u>: Number of cards in the 5 piles are 4, 1, 1, 1, 1.

$$\text{Number of ways } = \binom{8}{4} = 70$$

<u>Case 2</u>: Number of cards in the 5 piles are 3, 2, 1, 1, 1.

$$\text{Number of ways } = \binom{8}{3}\binom{5}{2} = 560$$

<u>Case 3</u>: Number of cards in the 5 piles are 2, 2, 2, 1, 1.

$$\text{Number of ways } = \frac{\binom{8}{2}\binom{6}{2}\binom{4}{2}}{3!} = 420$$

By Addition Principle, total number of ways to distribute the 8 distinct cards into 5 piles $= 70 + 560 + 420 = 1050$.

5(i)

Apply the Pigeonhole Principle twice:

Since there are 6 colours and only 5 jars, some jar must get pearls of two different colours. Since there are 11 pearls of a given colour and there are only 5 jars, some jar must contain at least $\left\lceil \frac{11}{5} \right\rceil = 3$ pearls of that colour. Since there are 6 colours and only 5 jars, some jar must get two pairs of same coloured pearls.

Illustration:

The figure below shows one possible distribution of pearls of 5 different colours (Pigeonhole Principle). Other distributions are permutations of the colours in the 5 jars.

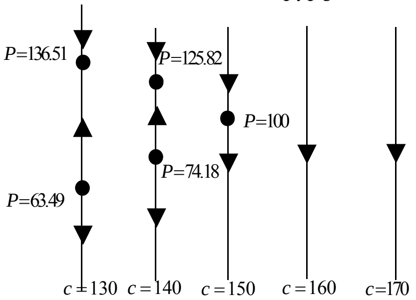
		Colour				
		1	**2**	**3**	**4**	**5**
Jar	**1**	≥3				
	2		≥3			
	3			≥3		
	4				≥3	
	5					≥3

By the Pigeonhole Principle, some jar must contain at least $\left\lceil \frac{11}{5} \right\rceil = 3$ pearls of the remaining colour. Thus the result follows.

(ii)	Number of ways of distributing 11 identical pearls into 4 identical bags is given by the number of partitions of 11 into 4 parts: 11 ways. 8+1+1+1, 7+2+1+1, 6+3+1+1, 6+2+2+1, 5+4+1+1, 5+3+2+1, 5+2+2+2, 4+4+2+1, 4+3+3+1, 4+3+2+2, 3+3+3+2. Alternatively, the number of partitions of 11 into 4 parts is equal to the number of partitions of 11 into parts the largest size of which is 4: 4+4+3, 4+4+2+1, 4+4+1+1+1, 4+3+3+1, 4+3+2+2, 4+3+2+1+1, 4+3+1+1+1+1, 4+2+2+2+1, 4+2+2+1+1+1, 4+2+1+1+1+1+1, 4+1+1+1+1+1+1+1. Since there are 6 possible colours to choose from, by the Multiplication Principle, number of ways of packing the remaining 11 pearls of the same colour into the 4 bags is equal to $6\times 11=66$.
(iii)	The number of ways of choosing 20 pearls if at least one pearl must be chosen from each jar is equal to the number of positive integer solutions to the equation $x_1+x_2+x_3+x_4+x_5=20$ **___(*)** where $1\le x_i\le 11$ for each $i=1,\ 2,\ 3,\ 4,\ 5$. Number of positive integer solutions to (*) is equal to $\binom{19}{4}=3876$. Number of positive integer solutions to (*) where $x_1\ge 12$ is $\binom{8}{4}=70$. By the Complementation Principle and Addition Principle, number of positive integer solutions to (*) where $1\le x_i\le 11$ for each $i=1,\ 2,\ 3,\ 4,\ 5$ is equal to $\binom{19}{4}-5\binom{8}{4}=3876-350=3526$.
(iv)	Let S be the event that pearls are chosen from the 5 jars and sewn onto 7 different pieces of jewellery such that only one pearl is sewn onto each piece of jewellery. Then $\lvert S\rvert=5^7$. Let A_i be the event that no pearl is chosen from the i-th jar, $i=1,\ 2,\ 3,\ 4,\ 5$. The number of ways a pearl can be sewn onto each of the 7 jewellery pieces, using at least one pearl from each of the 5 jars is given by $\left\lvert \overline{A}_1\cap\overline{A}_2\cap\overline{A}_3\cap\overline{A}_4\cap\overline{A}_5\right\rvert=\lvert S\rvert-\lvert A_1\cup A_2\cup A_3\cup A_4\cup A_5\rvert$. By the principle of inclusion and exclusion,

	$\left\|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5\right\| = \sum_{i=1}^{5}\left\|A_1\right\| - \sum_{i<j}\left\|A_i \cap A_j\right\| + \sum_{i<j<k}\left\|A_i \cap A_j \cap A_k\right\|$ $- \sum_{i<j<k<l}\left\|A_i \cap A_j \cap A_k \cap A_l\right\| + \left\|A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5\right\|$ $= \binom{5}{1}4^7 - \binom{5}{2}3^7 + \binom{5}{3}2^7 - \binom{5}{4}1^7 + 0$ Hence $\left\|\overline{A}_1 \cap \overline{A}_2 \cap \overline{A}_3 \cap \overline{A}_4 \cap \overline{A}_5\right\| = 5^7 - \binom{5}{1}4^7 + \binom{5}{2}3^7 - \binom{5}{3}2^7 + \binom{5}{4}1^7$.
6(i)(a)	Let $y = x^x$. $\ln y = x \ln x$ Diff. implicitly w.r.t x: $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x + x\left(\frac{1}{x}\right)$ $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = 1 + \ln x$ $\frac{\mathrm{d}y}{\mathrm{d}x} = y\left(1 + \ln x\right)$ $\therefore \frac{\mathrm{d}}{\mathrm{d}x}\left(x^x\right) = x^x\left(1 + \ln x\right)$ (shown)
(i)(b)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^x \mathrm{e}^{-x}\right) = x^x\left(-\mathrm{e}^{-x}\right) + \mathrm{e}^{-x}\left(x^x\left(1 + \ln x\right)\right)$ $= x^x \mathrm{e}^{-x} \ln x$ (shown)
(ii)	Let $\mathrm{f}\left(x_n, y_n\right) = x_n^{-x_n} - y_n \ln x_n$ Euler Formula: $y_{n+1} = y_n + h\,\mathrm{f}\left(x_n, y_n\right)$ $x_0 = 1, \quad y_0 = 0$ $x_1 = 1.2, \ y_1 = 0 + (0.2)\left(1^{-1} - 0\right) = 0.2$ $x_2 = 1.4, \ y_2 = 0.2 + (0.2)\left(1.2^{-1.2} - 0.2\ln 1.2\right) \approx 0.35341$ $= 0.353$ (3 s.f.)

(iii)	<table><tr><td>x</td><td>y</td><td>$x^{-x} - y\ln x$</td><td>y^*</td><td>$\frac{\Delta y}{\Delta x}$</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0.2</td><td>$\frac{1+0.767}{2}$</td></tr><tr><td>1.2</td><td>0.177</td><td>0.771</td><td>0.331</td><td>$\frac{0.771+0.513}{2} = \mathbf{0.642}$</td></tr><tr><td>1.4</td><td>**0.305**</td><td></td><td></td><td></td></tr></table>
(iv)	$\frac{dy}{dx} + y\ln x = x^{-x}$ ---(1) Integrating factor $e^{\int \ln x \, dx}$ $= e^{x\ln x - \int x\left(\frac{1}{x}\right) dx}$ $= e^{x\ln x - x}$ $= x^x e^{-x}$ Multiply $x^x e^{-x}$ to (1): $x^x e^{-x}\frac{dy}{dx} + yx^x e^{-x}\ln x = x^{-x}x^x e^{-x}$ $\Rightarrow \frac{d}{dx}\left(yx^x e^{-x}\right) = e^{-x}$ ___(*) $\Rightarrow yx^x e^{-x} = \int e^{-x} dx$ $= -e^{-x} + C$ Given that $y = 0$ when $x = 1$, we have $C = e^{-1}$. $\therefore y = \frac{e^{x-1} - 1}{x^x}$. When $x = 1.4$, $y = \frac{e^{1.4-1} - 1}{1.4^{1.4}} = 0.307$ (3 s.f.)
7(i)	a: growth rate b: carrying capacity of the pond c: harvesting (fishing) rate
(ii)	Consider $0.015P(200 - P) - c = 0$ $0.015P^2 - 3P + c = 0$ $P = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(0.015)c}}{2(0.015)}$ $= \frac{3 \pm \sqrt{9 - 0.06c}}{0.03}$ $= 100 \pm \frac{\sqrt{9 - 0.06c}}{0.03} = 100 \pm \sqrt{10000 - \frac{200}{3}c}$

	Bifurcation value : $c=\dfrac{9}{0.06}=150$ 
(iii)	$\dfrac{\mathrm{d}P}{\mathrm{d}t}=0.015P\left(200-P\right)$ $\displaystyle\int\frac{1}{P\left(200-P\right)}\,\mathrm{d}P=0.015\int 1\,\mathrm{d}t$ $\displaystyle\frac{1}{200}\int\frac{1}{P}+\frac{1}{200-P}\,\mathrm{d}P=0.015\int 1\,\mathrm{d}t$ $\displaystyle\frac{1}{200}\ln\left\|\frac{P}{200-P}\right\|=0.015t+D$ $\dfrac{P}{200-P}=A\mathrm{e}^{3t}$, where $A=\pm\mathrm{e}^{200D}$ When $t=0$, $P=120$: $A=1.5$ $P=\dfrac{300\mathrm{e}^{3t}}{1+1.5\mathrm{e}^{3t}}=\dfrac{300}{1.5+\mathrm{e}^{-3t}}$ As $t\to\infty$, $P=\dfrac{300}{1.5+\mathrm{e}^{-3t}}\to 200$ $\therefore k=200$
8(i)	$\dfrac{\mathrm{d}x}{\mathrm{d}t}=4x-6y$ $\Rightarrow\ \dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}=4\dfrac{\mathrm{d}x}{\mathrm{d}t}-6\dfrac{\mathrm{d}y}{\mathrm{d}t}$ $\Rightarrow\ \dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}=4\dfrac{\mathrm{d}x}{\mathrm{d}t}-6\left(8x-10y\right)$ $\Rightarrow\ \dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}=4\dfrac{\mathrm{d}x}{\mathrm{d}t}-48x+60y$ $\Rightarrow\ \dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}=4\dfrac{\mathrm{d}x}{\mathrm{d}t}-48x+10\left(4x-\dfrac{\mathrm{d}x}{\mathrm{d}t}\right)$ $\Rightarrow\ \dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}+6\dfrac{\mathrm{d}x}{\mathrm{d}t}+8x=0$ (shown)
(ii)	$\dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}+6\dfrac{\mathrm{d}x}{\mathrm{d}t}+8x=0$ ----(*) Characteristic equation: $m^2+6m+8=0$ $\therefore m=-2$ or $m=-4$

	General solution of (*) is $x = A\mathrm{e}^{-2t} + B\mathrm{e}^{-4t}$ $\frac{\mathrm{d}x}{\mathrm{d}t} = 4x - 6y$ $\Rightarrow y = \frac{1}{6}\left(4x - \frac{\mathrm{d}x}{\mathrm{d}t}\right)$ $\Rightarrow y = \frac{1}{6}\left(4\left(A\mathrm{e}^{-2t} + B\mathrm{e}^{-4t}\right) - \left(-2A\mathrm{e}^{-2t} - 4B\mathrm{e}^{-4t}\right)\right)$ $\Rightarrow y = \frac{1}{6}\left(6A\mathrm{e}^{-2t} + 8B\mathrm{e}^{-4t}\right)$ $\Rightarrow y = A\mathrm{e}^{-2t} + \frac{4}{3}B\mathrm{e}^{-4t}$
(iii)	Both species of animals will become extinct in the long run.